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Vellore Institute of Technology

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SCHOOL OF COMPUTER SCIENCE AND ENGINEERING (SCOPE)

LAB WEEK – 6

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SLOT: D2 + TD2

PROBLEM STATEMENT

Modern communities often face difficulties in identifying health risks at an early stage due to the absence of continuous monitoring and structured reporting systems. Most healthcare responses occur only after symptoms become severe or outbreaks spread within a locality. Traditional health monitoring methods rely heavily on manual reporting, hospital visits, or centralized medical systems, which delay early intervention.

Citizens generally lack a simple digital platform to record daily health conditions such as fever, respiratory symptoms, or general discomfort. At the same time, healthcare workers and local authorities do not have real-time visibility into emerging health patterns within communities.

This lack of timely data collection and monitoring increases the risk of delayed medical response, uncontrolled disease spread, and inefficient healthcare resource allocation.

Therefore, there is a need for a **Community Micro-Health Monitoring and Alert System** that enables citizens to report health parameters digitally, analyzes risk levels automatically, and notifies healthcare authorities for early preventive action.

OBJECTIVES

The primary objective of the proposed system is to design and develop a **web-based health monitoring platform** that facilitates early detection of potential health risks within a community.

Specific Objectives

1. To develop a user-friendly platform that allows citizens to submit daily health symptoms and basic medical parameters.
2. To implement a structured system for collecting and storing community health data securely.
3. To design a rule-based risk assessment mechanism that categorizes reported cases into:
    - Low Risk
    - Medium Risk
    - High Risk
4. To provide real-time alerts to health workers and administrators when abnormal health conditions are detected.

5. To enable administrators and doctors to monitor community health trends through dashboards and analytics.
6. To improve communication between citizens and healthcare authorities for faster intervention.
7. To create a scalable system capable of supporting smart city healthcare monitoring initiatives.

METHADOLOGY

The system adopts a **client–server architecture** where users interact through a web interface while data processing and storage are managed through cloud-based services.

The architecture consists of:

- Frontend Interface
- Backend Server
- Cloud Database
- Risk Analysis Module
- Alert Notification System

Workflow:

Citizen → Web Application → Database → Risk Analysis → Admin Alert

1 User Registration and Authentication

Two primary user roles are defined:

Citizen Module

Citizens register and log into the system to submit daily health information.

Admin/Health Worker Module

Administrators monitor submitted reports, analyze risks, and respond to alerts.

Authentication mechanisms ensure controlled system access.

2 Health Data Collection

Citizens submit health-related parameters such as:

- Body temperature
- Cough or respiratory symptoms
- General discomfort
- Environmental exposure indicators

The data is collected through structured web forms and transmitted securely to the database.

3 Database Management

A cloud-based database (Firebase Firestore) is used to store:

- User profiles
- Health reports
- Risk classifications
- Alert history

This enables real-time data availability and scalability.

4 Risk Assessment Mechanism

A rule-based evaluation algorithm processes submitted health parameters.

Example logic:

- Normal temperature → Low Risk
- Mild symptoms → Medium Risk
- High fever or multiple symptoms → High Risk

The system automatically assigns a risk level to each report.

5 Alert Generation System

When medium or high-risk conditions are detected:

1. The system generates alerts.

2. Notifications are displayed on the admin dashboard.
3. Health workers can review patient details and initiate action.

This supports early intervention and preventive healthcare management.

6 Dashboard and Monitoring Interface

Interactive dashboards provide:

- Real-time patient monitoring
- Risk distribution visualization
- Alert tracking
- Community health analytics

Administrators can observe trends and identify potential outbreaks.

7 System Implementation Technologies

The system is implemented using:

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Node.js with Express.js
- **Database:** Firebase Firestore
- **API Communication:** REST APIs
- **Deployment Environment:** Web-based platform

8 Testing and Validation

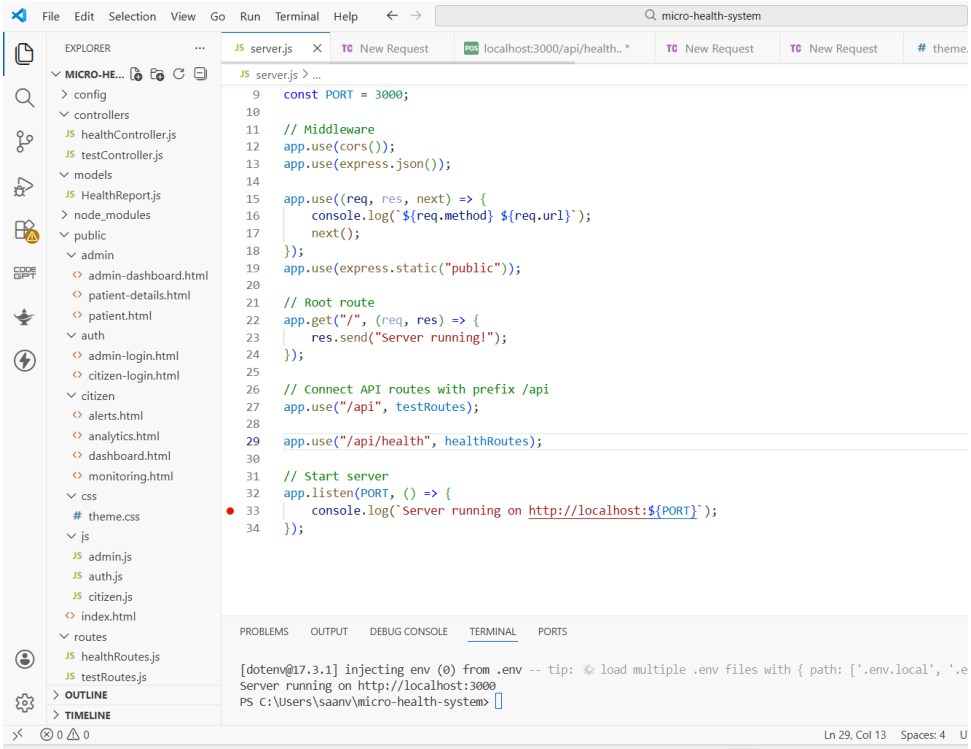
The system is tested by simulating multiple citizen health submissions to verify:

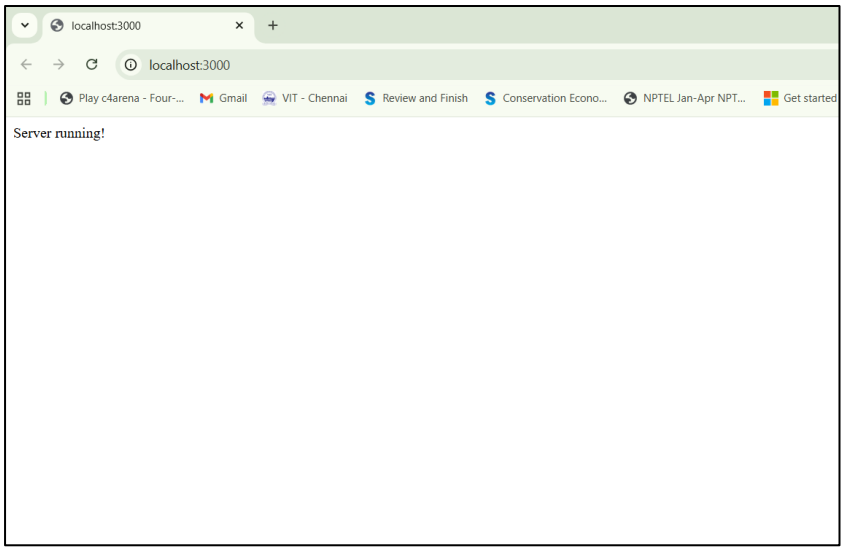
- Data storage accuracy
- Risk calculation correctness
- Alert generation functionality
- Dashboard updates

DEMO VIDEO LINK

https://drive.google.com/file/d/1goMgNlADwE-VFp6p6PYb7QxTi1yIvP6M/view?usp=sharing

SCREENSHOTS OF OUTPUT



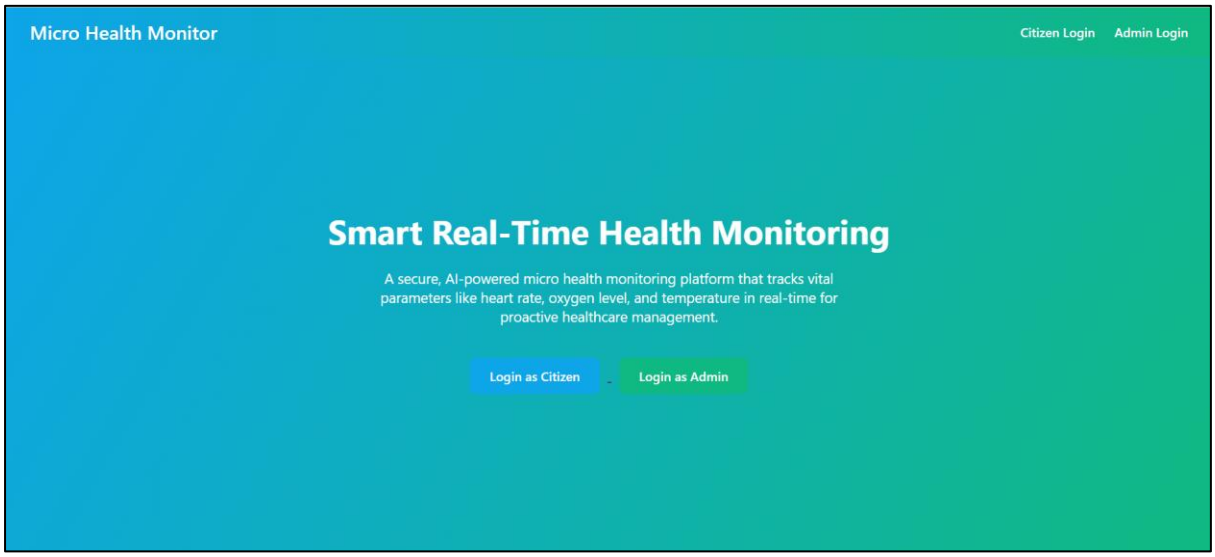


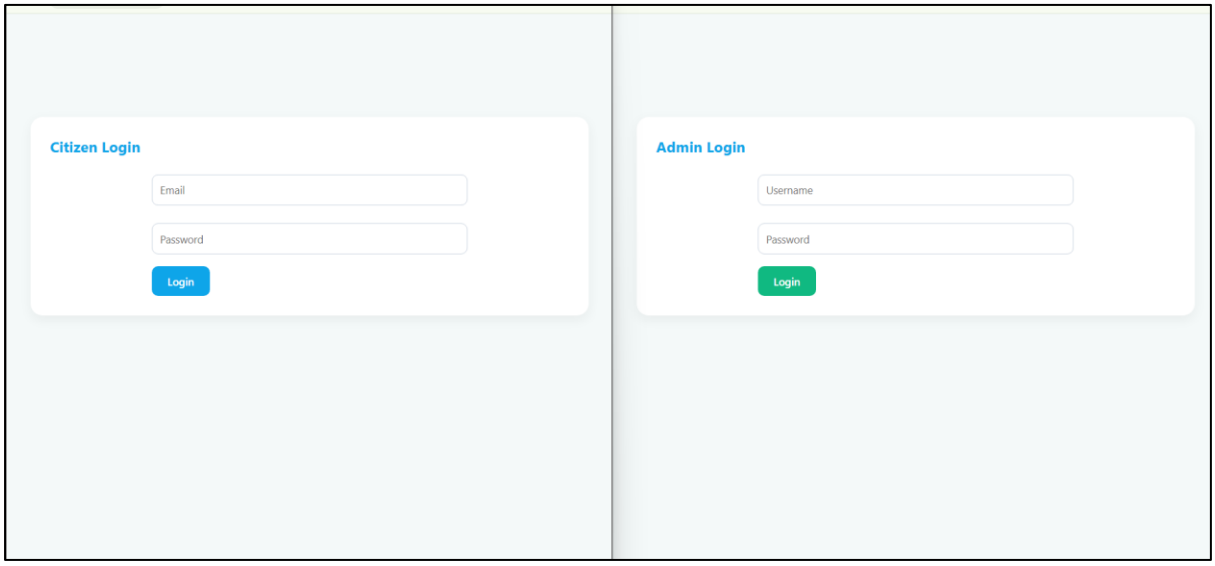
localhost:3000/api/health/alerts

Play c4arena - Four-... Gmail VIT - Chennai Review and Finish Conservation Econo... NPTEL Jan-Apr NPT... Get starte

Pretty print


```
[
  {
    "patient": "Patient A",
    "risk": "High",
    "time": "2026-03-02T04:12:16.406Z",
    "message": "High risk patient detected"
  }
]
```



Health Monitor

Dashboard Monitoring Alerts Logout

Alert History

Time	Type	Severity	Status
10:45 AM	Heart Rate	Critical	Resolved
Yesterday	Temperature	Warning	Checked

Health Monitor

Dashboard Monitoring Alerts Logout

Welcome Back 👋

❤️ Heart Rate

72 bpm

Normal

🌡 Temperature

36.8°C

Normal

🩸 Oxygen

98%

Good

📈 Status